

Don't Just Listen, Try Planning

Graph-based Retrieval-Generation Agent for
Long-form Audio Meeting Understanding

Quanwei Tang, Dong Zhang*, Shoushan Li, Guodong Zhou
NLP Lab, Soochow University, China

*dzhang@suda.edu.cn github.com/tangquanwei/GRGA

Motivation: Two Key Challenges in Long-form Audio Meeting QA

Challenge 1: Acoustic Missing

Speech LLMs often map audio → text tokens, losing **paralinguistic cues**: emotion, prosody, loudness, and tone.

Example: a “sudden loud voice at 19:00” is invisible to text-only representations.

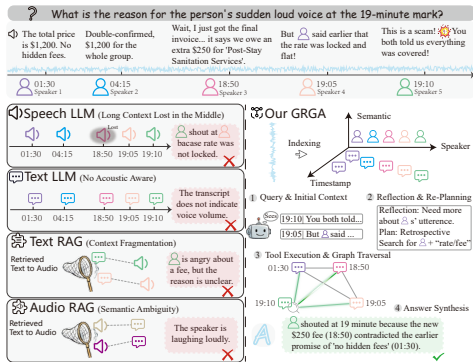
Challenge 2: Context Fragmentation

Long meetings exceed model context windows. Flat transcript RAG also struggles with **multi-hop evidence** across distant segments.

Example: linking a 1:30 description to an event at 19:00.

Our Solution

GRGA retrieves and reasons over a multi-dimensional graph;
LongAudioQA evaluates both text and acoustic grounding.



GRGA retrieves across semantic, speaker, and timestamp dimensions.

LongAudioQA: Benchmarking Long-form Audio Meeting QA

Three Sub-corpora

Dataset	Dur.	#Q&A
AliMeeting	14.9 h	3,013
AMI Meeting	18.2 h	3,243
DailyTalk	21.6 h	10,200

Five Question Types

Factual

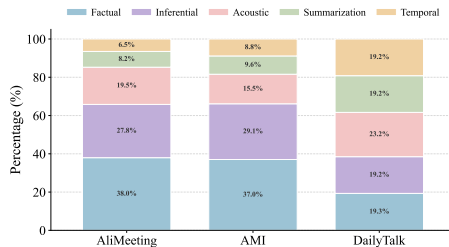
Inferential

Temporal

Summarization

Acoustic-Aware

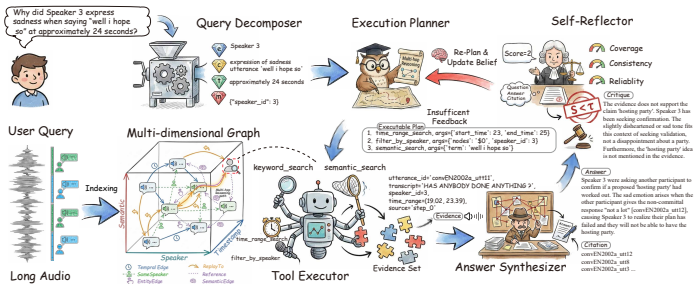
Quality control: hybrid LLM + human annotation;
 $\kappa=0.91$; timestamp IoU > 0.9.



Distribution across all three corpora.

Unique contribution: **Acoustic-Aware** questions require grounding textual semantics with paralinguistic signals.

GRGA Architecture: Multi-dimensional Graph + POMDP Planning



Meeting Graph $\mathcal{G}$

Nodes v_i : utterances with text, speaker, timestamp, and audio.

Edge types:

- e_{temp} : adjacent turns
- e_{reply} : replies within $\Delta t < 5s$
- e_{spk} : same-speaker links
- e_{ent} : shared keywords
- e_{sem} : LLM coreference chains

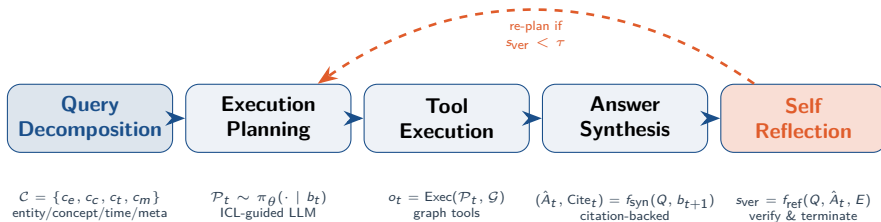
Agent as POMDP

$$\mathcal{M} = \langle S, \mathcal{A}, T, \mathcal{R}, \Omega, \mathcal{O}, \gamma \rangle$$

- Belief b_t : query + history
- Policy π : ICL-guided LLM
- Reward: Self-Reflector score

Pipeline: VAD $\rightarrow$ ASR $\rightarrow$ diarization $\rightarrow$ profile LLM.

GRGA Planning: A Cognitive “Search-Reason-Verify” Loop



Action Space (4 Graph Tools):

- **Semantic Search:** retrieve node text
- **Temporal Filter:** constrain time ranges
- **Graph Traversal:** follow speaker/semantic links
- **Audio Access:** inspect acoustic cues

Key Design Choices:

- **Training-free:** LLM policy via in-context learning
- **Adaptive depth:** shallow for factual, deeper for temporal/inferential
- **Self-correction:** reflection filters low-evidence answers

Experimental Results: GRGA Outperforms All Baselines

Category	Method	AliMeeting					AMI Meeting				
		Fact.	Infer.	Temp.	Summ.	Acou.	Fact.	Infer.	Temp.	Summ.	Acou.
Speech LLM	Qwen3-Omni	47.7	63.5	38.3	32.7	11.7	53.6	46.7	37.3	43.6	14.1
	MiMo-Audio	48.0	64.6	38.3	29.4	15.7	53.6	44.1	37.1	41.5	17.9
RAG	BGE-M3 (text)	43.2	29.7	31.6	15.4	18.5	32.6	24.6	23.8	25.4	25.3
	CLASP (audio)	25.4	19.5	18.9	14.8	16.3	24.1	20.3	22.8	11.6	14.8
Ours	GRGA	57.3	66.5	39.5	44.3	39.9	59.5	65.3	38.6	48.5	35.7

Key Findings:

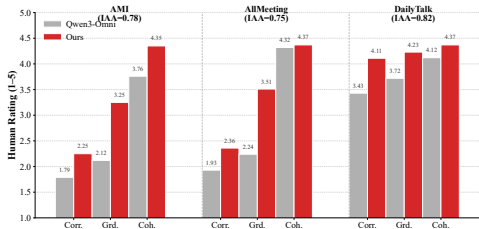
- **+10%** average gain over the best speech LLM on AMI.
- Inferential QA: **65.3%** vs. Text RAG **24.6%**.
- Acoustic QA: **39.9%** vs. next-best **18.5%**.

Why GRGA Wins:

- Graph structure preserves multi-hop meeting dependencies.
- Audio access supports acoustic-aware evidence.
- Reflection suppresses low-evidence answers.

Analysis: Human Evaluation & Ablation Study

Human Evaluation



Blind evaluation on 150 queries (3 annotators, IAA= 0.78).

+0.97 Groundedness confirms fewer hallucinations.

Ablation Study (AMI)

Setting	Avg. %	Δ
Full GRGA	49.49	—
w/o Query Planner	44.68	−4.80
w/o Reflection	39.64	−9.84
w/o Graph Traversal	38.21	−11.27
w/o Audio Access	34.71	−14.78
w/o Semantic Search	16.28	−33.21

Evidence Precision/Recall (AMI, $\pm 2s$)

Method	P	R
Qwen3-Omni	25.7	27.3
+ Text RAG	56.9	65.2
+ GRGA	76.2	69.7

Contributions

1. **LongAudioQA**: first large-scale long-form audio meeting QA benchmark with 16,456 Q&A pairs across three corpora.
2. **GRGA**: training-free agent that models heterogeneous audio as a multi-dimensional graph and plans iteratively.
3. **Evaluation**: automatic + human studies show consistent gains over Speech LLM and RAG baselines.

Limitations & Future Work

- Noisy ASR/diarization can still corrupt graph evidence.
- Iterative planning is slower than single-turn RAG.
- Future: streaming QA and broader long-form audio domains.

Long-form Audio QA

Acoustic Missing + Context Fragmentation



LongAudioQA Dataset

3 corpora, 5 types, 16K+ Q&A



GRGA Model

Multi-dim Graph + POMDP Agent



SOTA Performance

+10% on AMI, +19.3% Citation P

Code & Data:

github.com/tangquanwei/GRGA